The most important concept in elementary calculus is that of the limit. Recall that $\lim_{x \to x_0} f(x) = L$ intuitively means that values $f(x)$ of the function f can be made arbitrarily close to the real number L if values of x are chosen sufficiently close to, but not equal to, the real number x_0. In real analysis, the concepts of continuity, the derivative, and the definite integral were *all* defined using the concept of a limit. Complex limits play an equally important role in study of complex analysis. The concept of a complex limit is similar to that of a real limit in the sense that $\lim_{z \to z_0} f(z) = L$ will mean that the values $f(z)$ of the complex function f can be made arbitrarily close the complex number L if values of z are chosen sufficiently close to, but not equal to, the complex number z_0. Although outwardly similar, there is an important difference between these two concepts of limit. In a real limit, there are two directions from which x can approach x_0 on the real line, namely, from the left or from the right. In a complex limit, however, there are infinitely many directions from which z can approach z_0 in the complex plane. In order for a complex limit to exist, each way in which z can approach z_0 must yield the same limiting value.

In this section we will define the limit of a complex function, examine some of its properties, and introduce the concept of continuity for functions of a complex variable.

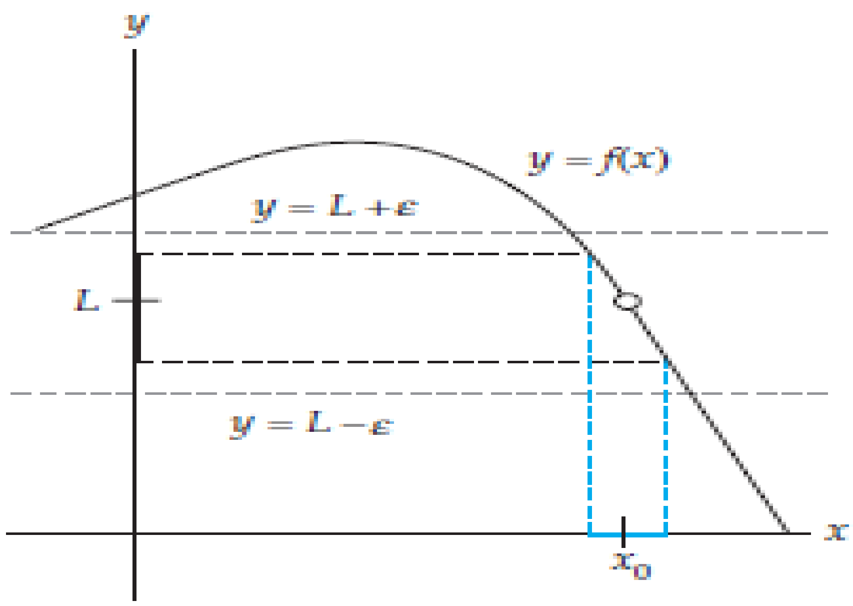
Limits and Continuity

Limit of a Real Function $f(x)$

The limit of f as x tends x_0 exists and is equal to L

if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ (1)

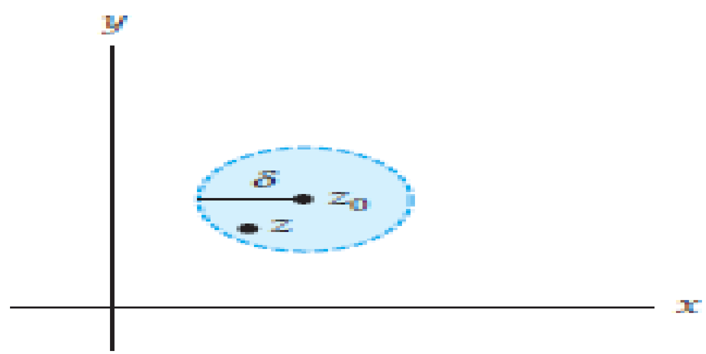
whenever $0 < |x - x_0| < \delta$.



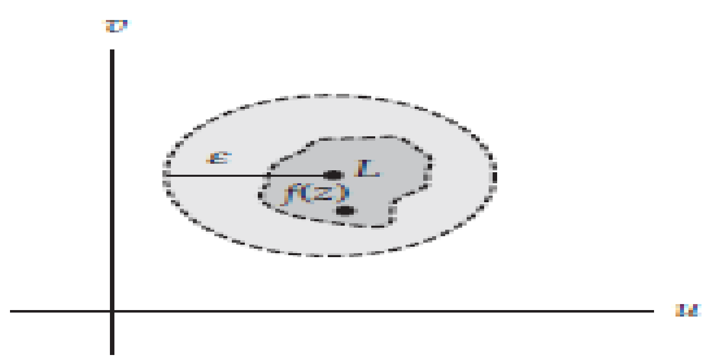
Complex Limits A complex limit is, in essence, the same as a real limit except that it is based on a notion of "close" in the complex plane. Because the distance in the complex plane between two points z_1 and z_2 is given by the modulus of the difference of z_1 and z_2, the precise definition of a complex limit will involve $|z_2 - z_1|$. For example, the phrase "$f(z)$ can be made arbitrarily close to the complex number L," can be stated precisely as: for every $\varepsilon > 0$, z can be chosen so that $|f(z) - L| < \varepsilon$. Since the modulus of a complex number is a *real* number, both ε and δ still represent small positive *real* numbers in the following definition of a complex limit. The complex analogue of (1) is:

Definition [redacted] Limit of a Complex Function

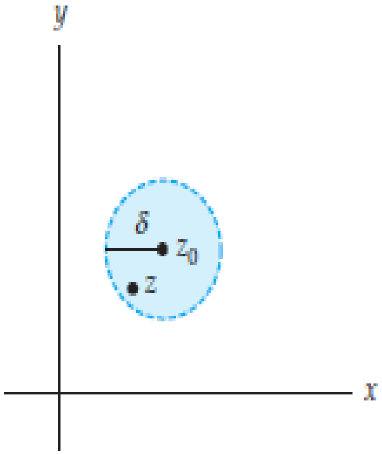
Suppose that a complex function f is defined in a deleted neighborhood of z_0 and suppose that L is a complex number. **The limit of f as z tends to z_0 exists and is equal to L**, written as $\lim_{z \to z_0} f(z) = L$, if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(z) - L| < \varepsilon$ whenever $0 < |z - z_0| < \delta$.



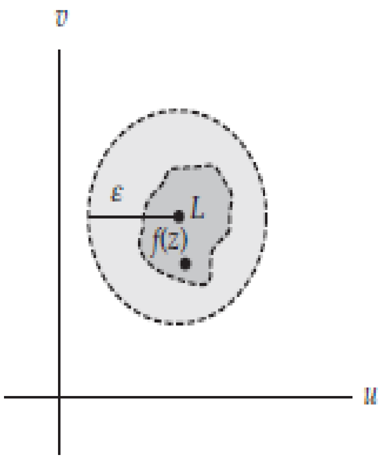
(a) Deleted δ-neighborhood of z_0



(b) ε-neighborhood of L



(a) Deleted δ-neighborhood of z_0

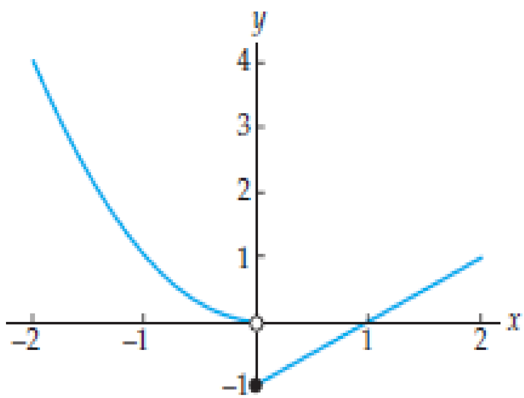


(b) ε-neighborhood of L

Because a complex function f has no graph, we rely on the concept of complex mappings to gain a geometric understanding of Definition . Recall that the set of points w in the complex plane satisfying $|w - L| < \varepsilon$ is called a neighborhood of L, and that this set consists of all points in the complex plane lying within, but not on, a circle of radius ε centered at the point L. Also recall that the set of points satisfying the inequalities $0 < |z - z_0| < \delta$ is called a deleted neighborhood of z_0 and consists of all points in the neighborhood $|z - z_0| < \delta$ excluding the point z_0. By Definition if $\lim_{z \to z_0} f(z) = L$ and if ε is any positive number, then there is a deleted neighborhood of z_0 of radius δ with the property that for every point z in this deleted neighborhood, $f(z)$ is in the ε neighborhood of L. That is, f maps the deleted neighborhood $0 < |z - z_0| < \delta$ in the z-plane into the neighborhood $|w - L| < \varepsilon$ in the w-plane. the deleted neighborhood of z_0 shown in color is mapped onto the set shown in dark gray in Figure 1(b). As required by Definition , the image lies within the ε-neighborhood of L shown in light gray in Figure (b).

Complex and real limits have many common properties, but there is at least one very important difference. For real functions, $\lim_{x \to x_0} f(x) = L$ if and only if $\lim_{x \to x_0^+} f(x) = L$ and $\lim_{x \to x_0^-} f(x) = L$. That is, there are two directions from which x can approach x_0 on the real line, from the right (denoted by $x \to x_0^+$) or from the left (denoted by $x \to x_0^-$). The real limit exists if and only if these two one-sided limits have the same value. For example, consider the real function defined by:

$$f(x) = \begin{cases} x^2, & x < 0 \\ x - 1, & x \geq 0 \end{cases}.$$



The limit of f does not exist as x approaches 0.

The limit of f as x approaches to 0 does not exist since $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} x^2 = 0$, but $\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} (x-1) = -1$.

For limits of complex functions, z is allowed to approach z_0 from *any* direction in the complex plane, that is, along any curve or path through z_0.

In order that $\lim_{z \to z_0} f(z)$ exists and equals L, we require that $f(z)$ approach the same complex number L along every possible curve through z_0.

Criterion for the Nonexistence of a Limit

If f approaches two complex numbers $L_1 \neq L_2$ for two different curves or paths through z_0, then $\lim_{z \to z_0} f(z)$ does not exist.

EXAMPLE 1 A Limit That Does Not Exist

Show that $\lim_{z \to 0} \frac{z}{\bar{z}}$ does not exist.

Solution We show that this limit does not exist by finding two different ways of letting z approach 0 that yield different values for $\lim_{z \to 0} \frac{z}{\bar{z}}$. First, we let z approach 0 along the real axis. That is, we consider complex numbers of the form $z = x + 0i$ where the real number x is approaching 0. For these points we have:

$$\lim_{z \to 0} \frac{z}{\bar{z}} = \lim_{x \to 0} \frac{x + 0i}{x - 0i} = \lim_{x \to 0} 1 = 1. \tag{2}$$

On the other hand, if we let z approach 0 along the imaginary axis, then $z = 0 + iy$ where the real number y is approaching 0. For this approach we have:

$$\lim_{z \to 0} \frac{z}{\bar{z}} = \lim_{y \to 0} \frac{0 + iy}{0 - iy} = \lim_{y \to 0} (-1) = -1. \tag{3}$$

Since the values in (2) and (3) are not the same, we conclude that $\lim_{z \to 0} \frac{z}{\bar{z}}$ does not exist.

Limit must be independent of direction approach

The limit $\lim_{z\to 0} \frac{z}{\bar{z}}$ from Example 1 did not exist because the values of $\lim_{z\to 0} \frac{z}{\bar{z}}$ as z approached 0 along the real and imaginary axes did not agree. However, even if these two values did agree, the complex limit may still fail to exist. In general, computing values of $\lim_{z\to z_0} f(z)$ as z approaches z_0 from different directions, as in Example 1, can prove that a limit *does not exist*, but this technique cannot be used to prove that a limit *does exist*. In order to prove that a limit does exist we must use Definition directly. This requires demonstrating that for every positive real number ε there is an appropriate choice of δ that meets the requirements of Definition Such proofs are commonly called "epsilon-delta proofs." Even for relatively simple functions, epsilon-delta proofs can be quite involved. Since this is an introductory text, we restrict our attention to what, in our opinion, are straightforward examples of epsilon-delta proofs.

EXAMPLE 2 An Epsilon-Delta Proof of a Limit

Prove that $\lim_{z \to 1+i} (2+i)z = 1+3i$.

Solution According to Definition $\lim_{z \to 1+i} (2+i)z = 1+3i$, if, for every $\varepsilon > 0$, there is a $\delta > 0$ such that $|(2+i)z - (1+3i)| < \varepsilon$ whenever $0 < |z-(1+i)| < \delta$. Proving that the limit exists requires that we find an appropriate value of δ for a given value of ε. In other words, for a given value of ε we must find a positive number δ with the property that if $0 < |z-(1+i)| < \delta$, then $|(2+i)z-(1+3i)| < \varepsilon$. One way of finding δ is to "work backwards." The idea is to start with the inequality:

$$|(2+i)z - (1+3i)| < \varepsilon \tag{4}$$

and then use properties of complex numbers and the modulus to manipulate this inequality until it involves the expression $|z-(1+i)|$. Thus, a natural first step is to factor $(2+i)$ out of the left-hand side of (4):

$$|2+i| \cdot \left| z - \frac{1+3i}{2+i} \right| < \varepsilon. \tag{5}$$

Because $|2+i| = \sqrt{5}$ and $\dfrac{1+3i}{2+i} = 1+i$, (5) is equivalent to:

$$\sqrt{5} \cdot |z-(1+i)| < \varepsilon \quad \text{or} \quad |z-(1+i)| < \frac{\varepsilon}{\sqrt{5}}. \tag{6}$$

Thus, (6) indicates that we should take $\delta = \varepsilon/\sqrt{5}$. Keep in mind that the choice of δ is not unique. Our choice of $\delta = \varepsilon/\sqrt{5}$ is a result of the particular algebraic manipulations that we employed to obtain (6). Having found δ we now present the formal proof that $\lim_{z \to 1+i} (2+i)z = 1+3i$ that does not indicate how the choice of δ was made:

Given $\varepsilon > 0$, let $\delta = \varepsilon/\sqrt{5}$. If $0 < |z-(1+i)| < \delta$, then we have $|z-(1+i)| < \varepsilon/\sqrt{5}$. Multiplying both sides of the last inequality by $|2+i| = \sqrt{5}$ we obtain:

$$|2+i| \cdot |z-(1+i)| < \sqrt{5} \cdot \frac{\varepsilon}{\sqrt{5}} \quad \text{or} \quad |(2+i)z-(1+3i)| < \varepsilon.$$

Therefore, $|(2+i)z-(1+3i)| < \varepsilon$ whenever $0 < |z-(1+i)| < \delta$. So, according to Definition , we have proven that $\lim_{z \to 1+i} (2+i)z = 1+3i$.

Limit of the Real Function $F(x, y)$

The limit of F as $(x,\ y)$ tends to $(x_0,\ y_0)$ exists and is equal to the real number L if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|F(x,\ y) - L| < \varepsilon$ whenever $0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$. (7)

The expression $\sqrt{(x - x_0)^2 + (y - y_0)^2}$ in (7) represents the distance between the points $(x,\ y)$ and $(x_0,\ y_0)$ in the Cartesian plane. Using (7), it is relatively easy to prove that:

$$\lim_{(x,y)\to(x_0,y_0)} 1 = 1, \quad \lim_{(x,y)\to(x_0,y_0)} x = x_0, \ \text{and} \lim_{(x,y)\to(x_0,y_0)} y = y_0. \tag{8}$$

If $\lim_{(x,y)\to(x_0,y_0)} F(x,y) = L$ and $\lim_{(x,y)\to(x_0,y_0)} G(x,y) = M$, then (7) can also be used to show:

$$\lim_{(x,y)\to(x_0,y_0)} cF(x,\ y) = cL, \quad c \text{ a real constant}, \tag{9}$$

$$\lim_{(x,y)\to(x_0,y_0)} (F(x,\ y) \pm G(x,\ y)) = L \pm M, \tag{10}$$

$$\lim_{(x,y)\to(x_0,y_0)} F(x,\ y) \cdot G(x,\ y) = L \cdot M, \tag{11}$$

and

$$\lim_{(x,y)\to(x_0,y_0)} \frac{F(x,\ y)}{G(x,\ y)} = \frac{L}{M}, \ M \neq 0. \tag{12}$$

Limits involving polynomial expressions in x and y can be easily computed using the limits in (8) combined with properties (9)–(12). For example,

$$\begin{aligned}\lim_{(x,y)\to(1,2)} (3xy^2 - y) &= 3\left(\lim_{(x,y)\to(1,2)} x\right)\left(\lim_{(x,y)\to(1,2)} y\right)\left(\lim_{(x,y)\to(1,2)} y\right) - \lim_{(x,y)\to(1,2)} y \\ &= 3 \cdot 1 \cdot 2 \cdot 2 - 2 = 10.\end{aligned}$$

In general, if $p(x,\ y)$ is a two-variable polynomial function, then (8)–(12) can be used to show that

$$\lim_{(x,y)\to(x_0,y_0)} p(x,\ y) = p(x_0,\ y_0). \tag{13}$$

If $p(x,\ y)$ and $q(x,\ y)$ are two-variable polynomial functions and $q(x_0,\ y_0) \neq 0$, then (13) and (12) give:

$$\lim_{(x,y)\to(x_0,y_0)} \frac{p(x,y)}{q(x,y)} = \frac{p(x_0,y_0)}{q(x_0,y_0)}. \tag{14}$$

Theorem Real and Imaginary Parts of a Limit

Suppose that $f(z) = u(x,\ y) + iv(x,\ y)$, $z_0 = x_0 + iy_0$, and $L = u_0 + iv_0$. Then $\lim_{z\to z_0} f(z) = L$ if and only if

$$\lim_{(x,y)\to(x_0,y_0)} u(x,\ y) = u_0 \quad \text{and} \quad \lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0.$$

EXAMPLE 3 Using Theorem to Compute a Limit

Use Theorem to compute $\lim_{z \to 1+i} (z^2 + i)$.

Solution Since $f(z) = z^2 + i = x^2 - y^2 + (2xy + 1)i$, we can apply Theorem with $u(x, y) = x^2 - y^2$, $v(x, y) = 2xy + 1$, and $z_0 = 1 + i$. Identifying $x_0 = 1$ and $y_0 = 1$, we find u_0 and v_0 by computing the two real limits:

$$u_0 = \lim_{(x,y) \to (1,1)} (x^2 - y^2) \quad \text{and} \quad v_0 = \lim_{(x,y) \to (1,1)} (2xy + 1).$$

Since both of these limits involve only multivariable polynomial functions, we can use (13) to obtain:

$$u_0 = \lim_{(x,y) \to (1,1)} (x^2 - y^2) = 1^2 - 1^2 = 0$$

and
$$v_0 = \lim_{(x,y) \to (1,1)} (2xy + 1) = 2 \cdot 1 \cdot 1 + 1 = 3,$$

and so $L = u_0 + iv_0 = 0 + i(3) = 3i$. Therefore, $\lim_{z \to 1+i} (z^2 + i) = 3i$.

Theorem Properties of Complex Limits

Suppose that f and g are complex functions. If $\lim_{z \to z_0} f(z) = L$ and $\lim_{z \to z_0} g(z) = M$, then

(i) $\lim_{z \to z_0} cf(z) = cL$, c a complex constant,

(ii) $\lim_{z \to z_0} (f(z) \pm g(z)) = L \pm M$,

(iii) $\lim_{z \to z_0} f(z) \cdot g(z) = L \cdot M$, and

(iv) $\lim_{z \to z_0} \dfrac{f(z)}{g(z)} = \dfrac{L}{M}$, provided $M \neq 0$.

$$\lim_{z \to z_0} c = c, \quad c \text{ a complex constant}, \tag{15}$$

$$\lim_{z \to z_0} z = z_0. \tag{16}$$

EXAMPLE 4 **Computing Limits with Theorem**

Use Theorem and the basic limits (15) and (16) to compute the limits

(a) $\lim_{z \to i} \dfrac{(3+i)z^4 - z^2 + 2z}{z+1}$

(b) $\lim_{z \to 1+\sqrt{3}i} \dfrac{z^2 - 2z + 4}{z - 1 - \sqrt{3}i}$

Solution

(a) By Theorem (*iii*) and (16), we have:

$$\lim_{z \to i} z^2 = \lim_{z \to i} z \cdot z = \left(\lim_{z \to i} z\right) \cdot \left(\lim_{z \to i} z\right) = i \cdot i = -1.$$

Similarly, $\lim_{z \to i} z^4 = i^4 = 1$. Using these limits, Theorems (*i*), (*ii*), and the limit in (16), we obtain:

$$\begin{aligned}\lim_{z \to i} \left((3+i)z^4 - z^2 + 2z\right) &= (3+i)\lim_{z \to i} z^4 - \lim_{z \to i} z^2 + 2\lim_{z \to i} z \\ &= (3+i)(1) - (-1) + 2(i) \\ &= 4 + 3i,\end{aligned}$$

and $\lim_{z \to i}(z+1) = 1 + i$. Therefore, by Theorem (*iv*), we have:

$$\lim_{z \to i} \frac{(3+i)z^4 - z^2 + 2z}{z+1} = \frac{\lim_{z \to i}\left((3+i)z^4 - z^2 + 2z\right)}{\lim_{z \to i}(z+1)} = \frac{4+3i}{1+i}.$$

After carrying out the division, we obtain $\lim_{z \to i} \dfrac{(3+i)z^4 - z^2 + 2z}{z+1} = \dfrac{7}{2} - \dfrac{1}{2}i.$

(b) In order to find $\displaystyle\lim_{z\to 1+\sqrt{3}i} \frac{z^2-2z+4}{z-1-\sqrt{3}i}$, we proceed as in (a):

$$\begin{aligned}\lim_{z\to 1+\sqrt{3}i}(z^2-2z+4) &= \left(1+\sqrt{3}i\right)^2 - 2\left(1+\sqrt{3}i\right)+4\\ &= -2+2\sqrt{3}i-2-2\sqrt{3}i+4=0,\end{aligned}$$

and $\displaystyle\lim_{z\to 1+\sqrt{3}i}(z-1-\sqrt{3}i) = 1+\sqrt{3}i-1-\sqrt{3}i = 0$. It appears that we cannot apply Theorem (*iv*) since the limit of the denominator is 0. However, in the previous calculation we found that $1+\sqrt{3}i$ is a root of the quadratic polynomial z^2-2z+4. recall that if z_1 is a root of a quadratic polynomial, then $z-z_1$ is a factor of the polynomial. Using long division, we find that

$$z^2-2z+4=\left(z-1+\sqrt{3}i\right)\left(z-1-\sqrt{3}i\right).$$

$$\lim_{z \to 1+\sqrt{3}i} \frac{z^2 - 2z + 4}{z - 1 - \sqrt{3}i} = \lim_{z \to 1+\sqrt{3}i} \frac{\left(z - 1 + \sqrt{3}i\right)\left(z - 1 - \sqrt{3}i\right)}{z - 1 - \sqrt{3}i}$$

$$= \lim_{z \to 1+\sqrt{3}i} \left(z - 1 + \sqrt{3}i\right).$$

By Theorem (*ii*) and the limits in (15) and (16), we then have

$$\lim_{z \to 1+\sqrt{3}i} \left(z - 1 + \sqrt{3}i\right) = 1 + \sqrt{3}i - 1 + \sqrt{3}i = 2\sqrt{3}i.$$

Therefore, $\displaystyle\lim_{z \to 1+\sqrt{3}i} \frac{z^2 - 2z + 4}{z - 1 - \sqrt{3}i} = 2\sqrt{3}i.$

Continuity of Real Functions Recall that if the limit of a real function f as x approaches the point x_0 exists and agrees with the value of the function f at x_0, then we say that f is *continuous* at the point x_0.

Continuity of a Real Function $f(x)$

A function f is continuous at a point x_0 if $\lim_{x \to x_0} f(x) = f(x_0)$. (17)

In real analysis, we visualize the concept of continuity using the graph of the function f. Informally, a function f is continuous if there are no breaks or holes in the graph of f. Since we cannot graph a complex function, our discussion of the continuity of complex functions will be primarily algebraic in nature.

Definition [redacted] Continuity of a Complex Function

A complex function f is continuous at a point z_0 if

$$\lim_{z \longrightarrow z_0} f(z) = f(z_0).$$

Criteria for Continuity at a Point

A complex function f is continuous at a point z_0 if each of the following three conditions hold:

(*i*) $\lim_{z \longrightarrow z_0} f(z)$ *exists,*

(*ii*) f *is defined at z_0, and*

(*iii*) $\lim_{z \longrightarrow z_0} f(z) = f(z_0)$.

If a complex function f is not continuous at a point z_0 then we say that f is discontinuous at z_0. For example, the function $f(z) = \dfrac{1}{1+z^2}$ is discontinuous at $z = i$ and $z = -i$.

EXAMPLE 5 Checking Continuity at a Point

Consider the function $f(z) = z^2 - iz + 2$. In order to determine if f is continuous at, say, the point $z_0 = 1 - i$, we must find $\lim_{z \to z_0} f(z)$ and $f(z_0)$, then check to see whether these two complex values are equal. From Theorem and the limits in (15) and (16) we obtain:

$$\lim_{z \to z_0} f(z) = \lim_{z \to 1-i} (z^2 - iz + 2) = (1-i)^2 - i(1-i) + 2 = 1 - 3i.$$

Furthermore, for $z_0 = 1 - i$ we have:

$$f(z_0) = f(1-i) = (1-i)^2 - i(1-i) + 2 = 1 - 3i.$$

Since $\lim_{z \to z_0} f(z) = f(z_0)$, we conclude that $f(z) = z^2 - iz + 2$ is continuous at the point $z_0 = 1 - i$.

Properties of Continuous Functions Because the concept of continuity is defined using the complex limit, various properties of complex limits can be translated into statements about continuity.

Continuity of a Real Function $F(x, y)$

A function F is continuous at a point $(x_0,\ y_0)$ if

$$\lim_{(x,y)\to(x_0,y_0)} F(x,\ y) = F(x_0,\ y_0).$$

Theorem Real and Imaginary Parts of a Continuous Function

Suppose that $f(z) = u(x,\ y) + iv(x,\ y)$ and $z_0 = x_0 + iy_0$. Then the complex function f is continuous at the point z_0 if and only if both real functions u and v are continuous at the point $(x_0,\ y_0)$.

EXAMPLE 7 **Checking Continuity Using Theorem**

Show that the function $f(z) = \bar{z}$ is continuous on $\mathbf{C}$.

Solution According to Theorem , $f(z) = \bar{z} = \overline{x + iy} = x - iy$ is continuous at $z_0 = x_0 + iy_0$ if both $u(x, y) = x$ and $v(x, y) = -y$ are continuous at (x_0, y_0). Because u and v are two-variable polynomial functions, it follows from (13) that:

$$\lim_{(x,y)\to(x_0,y_0)} u(x, y) = x_0 \quad \text{and} \quad \lim_{(x,y)\to(x_0,y_0)} v(x, y) = -y_0.$$

This implies that u and v are continuous at (x_0, y_0), and, therefore, that f is continuous at $z_0 = x_0 + iy_0$ by Theorem . Since $z_0 = x_0 + iy_0$ was an arbitrary point, we conclude that the function $f(z) = \bar{z}$ is continuous on $\mathbf{C}$.

Theorem Properties of Continuous Functions

If f and g are continuous at the point z_0, then the following functions are continuous at the point z_0:

(*i*) cf, c a complex constant,

(*ii*) $f \pm g$,

(*iii*) $f \cdot g$, and

(*iv*) $\dfrac{f}{g}$ provided $g(z_0) \neq 0$.

Polynomial functions are continuous on the entire complex plane C.

Continuity of Rational Functions

Rational functions are continuous on their domains.

(*i*) Analogous to real analysis, we can also define the concepts of infinite limits and limits at infinity for complex functions. Intuitively, the limit $\lim_{z\to\infty} f(z) = L$ means that values $f(z)$ of the function f can be made arbitrarily close to L if values of z are chosen so that $|z|$ is sufficiently large. A precise statement of a **limit at infinity** is:

The limit of f as z tends to ∞ exists and is equal to L if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(z) - L| < \varepsilon$ whenever $|z| > 1/\delta$.

Using this definition it is not hard to show that:

$$\lim_{z\to\infty} f(z) = L \text{ if and only if } \lim_{z\to 0} f\left(\frac{1}{z}\right) = L.$$

Similarly, the **infinite limit** $\lim_{z\to z_0} f(z) = \infty$ is defined by:

The limit of f as z tends to z_0 is ∞ if forevery $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(z)| > 1/\varepsilon$ whenever $0 < |z - z_0| < \delta$.

From this definition we obtain the following result:

$$\lim_{z\to z_0} f(z) = \infty \text{ if and only if } \lim_{z\to z_0} \frac{1}{f(z)} = 0.$$

Show that the following limits are true

$$\lim_{z \to z_0} (az + c) = az_0 + c$$

$$|f(z) - (az_0 + c)| < \varepsilon$$

$$|az + c - az_0 - c| = |a|\,|z - z_0| < \varepsilon$$

$$|z - z_0| < \frac{\varepsilon}{|a|}$$

$\delta = \frac{\varepsilon}{|a|}$; so, the required is obtained

$$\lim_{z \to z_0} (z^2+c) = z_0^2 + c$$

$$|f(z) - (z_0^2+c)| = |z^2+c-z_0^2-c| = |z^2-z_0^2|$$

$$|z^2-z_0^2| = |z-z_0|\,|z+z_0|$$

Let $|z-z_0| < 1$;

if so, $|z+z_0| = |z-z_0+2z_0| \leq |z-z_0| + |2z_0| <$

$$< 1 + 2|z_0|$$

$$|f(z) - w_0| = |z-z_0|\,|z+z_0| < |z-z_0|(1+2|z_0|) < \varepsilon$$

$$|z-z_0| < \frac{\varepsilon}{1+2|z_0|}$$

if $\delta = \min\left\{1, \frac{\varepsilon}{1+2|z_0|}\right\}$;

the required is obtained.

Show that

$$\lim_{z \to z_0} \operatorname{Re} z = \operatorname{Re} z_0$$ is true.

Because

$|\operatorname{Re} z| \leq |z|$;

$$|f(z) - w_0| = |\operatorname{Re} z - \operatorname{Re} z_0| = |\operatorname{Re}(z - z_0)| \leq |z - z_0| < \varepsilon$$

$\delta = \varepsilon$ is choosen.

The required is obtained.

$$\lim_{z \to 2i} (2x+iy^2) = 4i \quad ?$$

$0 < |z-2i| < \delta \qquad |2x+iy^2-4i| < \varepsilon$

$|2x+i(y^2-4)| < \varepsilon$

$|2x+i(y^2-4)| \leq |2x| + |y^2-4| = 2|x| + |y-2||y+2|$

$|2x+iy^2-4i| < \varepsilon$ is satisfied while

$2|x| < \frac{\varepsilon}{2}$ and

$|y-2||y+2| < \frac{\varepsilon}{2}$

$\to |x| < \frac{\varepsilon}{4}$

$\to |y+2| = |y-2+4|$

$\leq |y-2| + 4$

when $|y-2| < 1$;

$|y+2| < 5$

Thus, $|y-2| < \min\left(\frac{\varepsilon}{10}, 1\right)$

If $\delta = \min\left(\frac{\varepsilon}{10}, 1\right)$, $0 \leq |z-2i| < \delta$, $|f(z)-4i| < \varepsilon$

is satisfied.

Really,

$|z-2i| < \delta \qquad |x+iy-2i| < \delta$

$|x+i(y-2)| < \delta$

$|x| < |x+i(y-2)|$

$|y-2| < |x+i(y-2)|$

$|x| < \delta < \frac{\varepsilon}{10} < \frac{\varepsilon}{4}$

$|y-2| < \delta < 1 \qquad |y+2| < 5$

$|y-2| < \delta < \frac{\varepsilon}{10} \qquad |y-2| < \frac{\varepsilon}{10}$

$|f(z)-4i| = |2x+i(y^2-4)| \leq 2|x| + |y-2||y+2| < 2 \cdot \frac{\varepsilon}{4} + \frac{\varepsilon}{10} \cdot 5 = \varepsilon$

if $f(z)=\frac{1}{z}$ $\lim\limits_{z\to i} f(z)=-i$?

for $\varepsilon>0$, $\exists\, \delta>0$, $0<|z-i|<\delta$, $\left|\frac{1}{z}+i\right|<\varepsilon$

$$\left|\frac{1}{z}+i\right|=\left|\frac{1+iz}{z}\right|=\frac{|1+iz|}{|z|}=\frac{|i|\,|z-i|}{|z|}=\frac{|z-i|}{|z|}=\frac{\delta}{1-\delta}$$

if $\delta=\frac{\varepsilon}{1+\varepsilon}$ is selected, the required is obtained.

If f has limit at z_0, it is unique.

Proof: Let us assume $\lim_{z \to z_0} f(z) = a$ $\lim_{z \to z_0} f(z) = b$

and $a \neq b$

Let $2\varepsilon = |a-b|$

$2\varepsilon = |a-b| = |a - f(z) + f(z) + b| \leq |\underbrace{f(z)-a}_{\varepsilon}| + |\underbrace{f(z)-b}_{\varepsilon}| < 2\varepsilon$

$2\varepsilon < 2\varepsilon$ is not possible ; $\underline{\underline{a=b}}$

$$f(z) = \begin{cases} \dfrac{z^4 - z_0^4}{z - z_0} & z \neq z_0 \\ 4 \cdot z^3 & z = z_0 \end{cases}$$

İf the function is continuous at z_0

$$z^4 - z_0^4 = z_0^4 \left(\left(\frac{z}{z_0} \right)^4 - 1 \right) = z_0^4 \left(w^4 - 1 \right) = z_0^4 \left((w-1)(w^3 + w^2 + w + 1) \right)$$

$$z - z_0 = z_0 \left(\frac{z}{z_0} - 1 \right) = (w - 1) . z_0$$

$$w^4 - 1 = (w - 1) . (w^3 + w^2 + w + 1)$$

$$\begin{aligned} x^4 - y^4 &= (x^2 - y^2) . (x^2 + y^2) \\ &= (x - y)(x + y)(x^2 + y^2) \\ &= (x - y)(x^3 + xy^2 + yx^2 + y^3) \end{aligned}$$

$$\frac{z^4 - z_0^4}{z - z_0} = z_0^3 \cdot (w^3 + w^2 + w + 1) \qquad \text{while } z \to z_0 ;\ w \to 1$$

$$\lim_{z \to z_0} f(z) = \lim_{w \to 1} z_0^3 (w^3 + w^2 + w + 1) = 4 z_0^3$$

İt is continuous

$$f(z) = \frac{\bar{z} - \bar{z}_0}{z - z_0} \qquad z_0 \neq 0$$

is given

$$z - z_0 = t e^{i\alpha}$$

$$\bar{z} - \bar{z}_0 = t \cdot e^{-i\alpha} \implies \frac{\bar{z} - \bar{z}_0}{z - z_0} = e^{-2i\alpha}$$

Limit of the function does not exist as $z \to z_0$ Because of α

Compute the following limits if they exist:

(a) $\lim\limits_{z \to -i} \dfrac{iz^3+1}{z^2+1}$;

(b) $\lim\limits_{z \to \infty} \dfrac{4+z^2}{(z-1)^2}$;

(c) $\lim\limits_{z \to 0} \dfrac{\operatorname{Im}(z)}{z}$.

Solution. (a)

$$\begin{aligned}
\lim_{z \to -i} \frac{iz^3+1}{z^2+1} &= \lim_{z \to -i} \frac{i(z^3+i^3)}{z^2+1} \\
&= \lim_{z \to -i} \frac{i(z+i)(z^2-iz+i^2)}{(z+i)(z-i)} \\
&= \lim_{z \to -i} \frac{i(z^2-iz+i^2)}{z-i} \\
&= i\frac{\lim_{z \to -i}(z^2-iz+i^2)}{\lim_{z \to -i}(z-i)} \\
&= i\frac{\lim_{z \to -i} z^2 - i\lim_{z \to -i} z + \lim_{z \to -i} i^2}{\lim_{z \to -i} z - \lim_{z \to -i} i} \\
&= \frac{i((-i)^2 - i(-i) + i^2)}{-i-i} = \frac{3}{2}
\end{aligned}$$

(b)

$$\begin{aligned}
\lim_{z\to\infty}\frac{4+z^2}{(z-1)^2} &= \lim_{z\to 0}\frac{4+z^{-2}}{(z^{-1}-1)^2} \\
&= \lim_{z\to 0}\frac{4z^2+1}{(1-z)^2} = \frac{\lim_{z\to 0}(4z^2+1)}{\lim_{z\to 0}(1-z)^2} \\
&= \frac{4(\lim_{z\to 0} z)^2+\lim_{z\to 0} 1}{(\lim_{z\to 0} 1-\lim_{z\to 0} z)^2} = 1
\end{aligned}$$

(c) Since

$$\lim_{\mathrm{Re}(z)=0, z\to 0}\frac{\mathrm{Im}(z)}{z} = \lim_{y\to 0}\frac{y}{yi} = -i$$

and

$$\lim_{\mathrm{Im}(z)=0, z\to 0}\frac{\mathrm{Im}(z)}{z} = \lim_{x\to 0}\frac{0}{x} = 0$$

the limit does not exist.

Show the following limits:

(a) $\lim_{z\to\infty} \frac{4z^5}{z^5-42z} = 4;$

(b) $\lim_{z\to\infty} \frac{z^4}{z^2+42z} = \infty;$

(c) $\lim_{z\to\infty} \frac{(az+b)^3}{(cz+d)^3} = \frac{a^3}{c^3}$, if $c \neq 0$.

Solution. (a)

$$\lim_{z\to\infty} \frac{4z^5}{z^5-42z} = \lim_{z\to 0} \frac{4\left(\frac{1}{z}\right)^5}{\left(\frac{1}{z}\right)^5 - 42\left(\frac{1}{z}\right)} = \lim_{z\to 0} \frac{4}{1-42z^4} = 4.$$

(b) Notice that

$$\begin{aligned} \lim_{z\to\infty} \frac{z^4}{z^2+42z} \quad &\Longleftrightarrow \quad \lim_{z\to 0} \left[\frac{1/z^4}{1/z^2+42/z}\right]^{-1} \\ &\Longleftrightarrow \quad \lim_{z\to 0} \left[\frac{1}{z^2+42z^3}\right]^{-1} \\ &\Longleftrightarrow \quad \lim_{z\to 0} \left(z^2+42z^3\right) = 0. \end{aligned}$$

Therefore, $\lim_{z\to\infty} \frac{z^4}{z^2+42z} = \infty.$

(c)

$$\lim_{z\to\infty} \frac{(az+b)^3}{(cz+d)^3} = \lim_{z\to 0} \frac{(a/z+b)^3}{(c/z+d)^3} = \lim_{z\to 0} \frac{(a+bz)^3}{(c+dz)^3} = \frac{a^3}{c^3}.$$

Show that if $f(z)$ is continuous at z_0, so is $|f(z)|$.

Proof. Let $f(z) = u(x,y) + iv(x,y)$. Since $f(z)$ is continuous at $z_0 = x_0 + y_0 i$, $u(x,y)$ and $v(x,y)$ are continuous at (x_0, y_0). Therefore,

$$(u(x,y))^2 + (v(x,y))^2$$

is continuous at (x_0, y_0) since the sums and products of continuous functions are continuous. It follows that

$$|f(z)| = \sqrt{(u(x,y))^2 + (v(x,y))^2}$$

is continuous at z_0 since the compositions of continuous functions are continuous.

Ex. $f(z)=\begin{cases} \dfrac{z.\text{Re}z}{|z|} & , z\neq 0 \\ 0 & , z=0 \end{cases}$

Show that $f(z)$ is continous at $z_0=0$

1) For, $\forall\ \varepsilon>0$, there exists $\exists\ \delta>0$ such that

$|f(z)-f(z_0)|<\varepsilon$ is satisfied

while $|z-z_0|<\delta$. ??

$$|f(z)-f(0)|=\left|\frac{z.\text{Re}z}{|z|}-0\right|=\frac{|z|.|\text{Re}z|}{|z|}=|\text{Re}z|$$

Since $|\text{Re}z|\leq|z|$; if δ is chosen as ε; th

That is $\delta=\varepsilon$; $|f(z)-f(z_0)|<\varepsilon$ can be

while $|z-z_0|<\delta$; satisfied

2) $f(z_0)=f(0)=0$

3) $\lim\limits_{z\to 0} f(z)=0=f(0)$, so, $f(z)$ is continuous

Ex. Determine whether $f(z) = \dfrac{z^2+4z}{z^2+1}$ continuous at $z_0 = i$

$g(z) = z^2+4z$ and $h(z) = z^2+1$ are continuous in ~~whole~~ entire complex plane. Thus, f is continuous except points that are satisfying $h(z) = z^2+1 = 0$

Ex $\lim_{z \to z_0} z^2 = z_0^2$

For $\forall \varepsilon > 0$, there exist $\exists \delta > 0$, $|z^2 - z_0^2| < \varepsilon$ is satisfied

while $|z - z_0| < \delta$??

$$
\begin{aligned}
|z^2 - z_0^2| &= |z - z_0||z + z_0| \\
&= |z - z_0||z + 2z_0 - z_0| \\
&\leq |z - z_0|(|z - z_0| + |2z_0|)
\end{aligned}
$$

while $|z - z_0| < 1$

$$|z^2 - z_0^2| < |z - z_0|(1 + |2z_0|)$$

if we choose $\delta_1 = \dfrac{\varepsilon}{1 + 2|z_0|}$; $\delta = \min\left\{1, \dfrac{\varepsilon}{1 + 2|z_0|}\right\}$.

for $\forall \varepsilon > 0$; The requested δ has been found.

Ex: $R = \{z : |z| \leq 3\}$ is given. Determine whether $f(z) = z^2$ is continuous.

$$|z^2 - z_0^2| = |z - z_0||z + z_0|$$
$$\leq |z - z_0|(|z| + |z_0|)$$
$$< \underbrace{|z - z_0|}_{\Downarrow} \cdot 6$$

if we choose $\delta = \frac{\varepsilon}{6}$

$\Rightarrow |z^2 - z_0^2| \leq \varepsilon$ is found.

Here, δ depends on only ε.

So, $f(z)$ is smooth continuous function.

In real analysis we visualize a continuous function as a function whose graph has no breaks or holes in it. It is natural to ask if there is an analogous property for continuous complex functions. The answer is yes, but this property must be stated in terms of complex mappings. We begin by recalling that a parametric curve defined by parametric equations $x = x(t)$ and $y = y(t)$ is called continuous if the real functions x and y are continuous. In a similar manner, we say that a complex parametric curve defined by $z(t) = x(t) + iy(t)$ is **continuous** if both $x(t)$ and $y(t)$ are continuous real functions. As with parametric curves in the Cartesian plane, a continuous parametric curve in the complex plane has no breaks or holes in it. Such curves provide a means to visualize continuous complex functions.

If a complex function f is continuous on a set S, then the image of every continuous parametric curve in S must be a continuous curve.

Derivative of Complex Function

Suppose the complex function f is defined in a neighborhood of a point z_0. The derivative of f at z_0, denoted by $f'(z_0)$, is

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} \tag{1}$$

provided this limit exists.

If the limit in (1) exists, then the function f is said to be **differentiable** at z_0.

EXAMPLE 1 Using Definition

Use Definition to find the derivative of $f(z) = z^2 - 5z$.

Solution Because we are going to compute the derivative of f at any point, we replace z_0 in (1) by the symbol z. First,

$$f(z + \Delta z) = (z + \Delta z)^2 - 5(z + \Delta z) = z^2 + 2z\Delta z + (\Delta z)^2 - 5z - 5\Delta z.$$

Second,

$$\begin{aligned} f(z + \Delta z) - f(z) &= z^2 + 2z\Delta z + (\Delta z)^2 - 5z - 5\Delta z - (z^2 - 5z) \\ &= 2z\Delta z + (\Delta z)^2 - 5\Delta z. \end{aligned}$$

Then, finally, (1) gives

$$\begin{aligned} f'(z) &= \lim_{\Delta z \to 0} \frac{2z\Delta z + (\Delta z)^2 - 5\Delta z}{\Delta z} \\ &= \lim_{\Delta z \to 0} \frac{\Delta z(2z + \Delta z - 5)}{\Delta z} \\ &= \lim_{\Delta z \to 0} (2z + \Delta z - 5). \end{aligned}$$

The limit is $f'(z) = 2z - 5$.

Rules of Differentiation

The familiar rules of differentiation in the calculus of real variables carry over to the calculus of complex variables. "

Differentiation Rules

Constant Rules: $$\frac{d}{dz}c = 0 \text{ and } \frac{d}{dz}cf(z) = cf'(z) \tag{2}$$

Sum Rule: $$\frac{d}{dz}[f(z) \pm g(z)] = f'(z) \pm g'(z) \tag{3}$$

Product Rule: $$\frac{d}{dz}[f(z)g(z)] = f(z)g'(z) + f'(z)g(z) \tag{4}$$

Quotient Rule: $$\frac{d}{dz}\left[\frac{f(z)}{g(z)}\right] = \frac{g(z)f'(z) - f(z)g'(z)}{[g(z)]^2} \tag{5}$$

Chain Rule: $$\frac{d}{dz}f(g(z)) = f'(g(z))\, g'(z). \tag{6}$$

The power rule for differentiation of powers of z is also valid:

$$\frac{d}{dz}z^n = nz^{n-1}, \quad n \text{ an integer.} \tag{7}$$

Combining (7) with (6) gives the power rule for functions:

$$\frac{d}{dz}[g(z)]^n = n[g(z)]^{n-1}g'(z), \quad n \text{ an integer.} \tag{8}$$

EXAMPLE 2 Using the Rules of Differentiation

Differentiate:

(a) $f(z) = 3z^4 - 5z^3 + 2z$ (b) $f(z) = \dfrac{z^2}{4z+1}$ (c) $f(z) = \left(iz^2 + 3z\right)^5$

Solution

(**a**) Using the power rule (7), the sum rule (3), along with (2), we obtain

$$f'(z) = 3 \cdot 4z^3 - 5 \cdot 3z^2 + 2 \cdot 1 = 12z^3 - 15z^2 + 2.$$

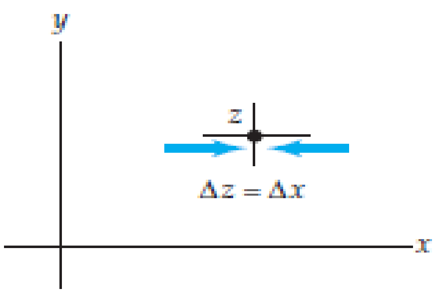
(**b**) From the quotient rule (5),

$$f'(z) = \frac{(4z+1) \cdot 2z - z^2 \cdot 4}{(4z+1)^2} = \frac{4z^2 + 2z}{(4z+1)^2}.$$

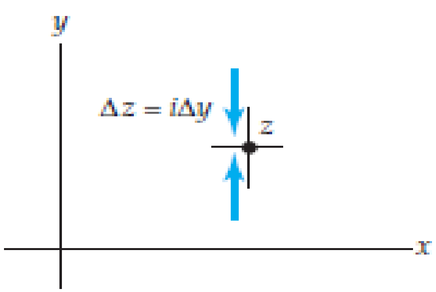
(**c**) In the power rule for functions (8) we identify $n = 5$, $g(z) = iz^2 + 3z$, and $g'(z) = 2iz + 3$, so that

$$f'(z) = 5(iz^2 + 3z)^4(2iz + 3).$$

For a complex function f to be differentiable at a point z_0, we know from the preceding chapter that the limit $\lim_{\Delta z \to 0} \dfrac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$ must exist and equal the same complex number from any direction; that is, the limit must exist regardless how Δz approaches 0. This means that in complex analysis, the requirement of differentiability of a function $f(z)$ at a point z_0 is a far greater demand than in real calculus of functions $f(x)$ where we can approach a real number x_0 on the number line from only two directions.



(a) $\Delta z \to 0$ along a line parallel to x-axis



(b) $\Delta z \to 0$ along a line parallel to y-axis

EXAMPLE 3 A Function That Is Nowhere Differentiable

Show that the function $f(z) = x + 4iy$ is not differentiable at any point z.

Solution Let z be any point in the complex plane. With $\Delta z = \Delta x + i\Delta y$,

$$f(z + \Delta z) - f(z) = (x + \Delta x) + 4i(y + \Delta y) - x - 4iy = \Delta x + 4i\Delta y$$

and so
$$\lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \to 0} \frac{\Delta x + 4i\Delta y}{\Delta x + i\Delta y}. \tag{9}$$

Now, as shown in Figure 1(a), if we let $\Delta z \to 0$ along a line parallel to the x-axis, then $\Delta y = 0$ and $\Delta z = \Delta x$ and

$$\lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \to 0} \frac{\Delta x}{\Delta x} = 1. \tag{10}$$

On the other hand, if we let $\Delta z \to 0$ along a line parallel to the y-axis as shown in Figure [illegible](b), then $\Delta x = 0$ and $\Delta z = i\Delta y$ so that

$$\lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \to 0} \frac{4i\Delta y}{i\Delta y} = 4. \tag{11}$$

In view of the obvious fact that the values in (10) and (11) are different, we conclude that $f(z) = x + 4iy$ is nowhere differentiable; that is, f is not differentiable at any point z.

Let

$$f(z) = \begin{cases} \bar{z}^3/z^2 & \text{if } z \neq 0 \\ 0 & \text{if } z = 0 \end{cases}$$

Show that

(a) $f(z)$ is continuous everywhere on $\mathbb{C}$;

(b) the complex derivative $f'(0)$ does not exist.

Proof. Since both $\bar{z}^3$ and z^2 are continuous on $\mathbb{C}^* = \mathbb{C}\setminus\{0\}$ and $z^2 \neq 0$, $f(z) = \bar{z}^3/z^2$ is continuous on $\mathbb{C}^*$.
At $z = 0$, we have

$$\lim_{z\to 0} |f(z)| = \lim_{z\to 0} \left|\frac{\bar{z}^3}{z^2}\right| = \lim_{z\to 0} |z| = 0$$

and hence $\lim_{z\to 0} f(z) = 0 = f(0)$. So f is also continuous at 0 and hence continuous everywhere on $\mathbb{C}$.
The complex derivative $f'(0)$, if exists, is given by

$$\lim_{z\to 0} \frac{f(z) - f(0)}{z - 0} = \lim_{z\to 0} \frac{\bar{z}^3}{z^3}.$$

Let $z = x + yi$. If $y = 0$ and $x \to 0$, then

$$\lim_{z=x\to 0} \frac{\bar{z}^3}{z^3} = \lim_{x\to 0} \frac{x^3}{x^3} = 1.$$

On the other hand, if $x = 0$ and $y \to 0$, then

$$\lim_{z=yi\to 0} \frac{\bar{z}^3}{z^3} = \lim_{x\to 0} \frac{(-yi)^3}{(yi)^3} = -1.$$

So the limit $\lim_{z\to 0} \bar{z}^3/z^3$ and hence $f'(0)$ do not exist.

Analytic Functions

A complex function $w = f(z)$ is said to be analytic at a point z_0 if f is differentiable at z_0 and at every point in some neighborhood of z_0.

A function f is analytic in a domain D if it is analytic at every point in D. The phrase "analytic *on* a domain D" is also used.

Entire Functions

A function that is analytic at every point z in the complex plane is said to be an **entire function**. In view of differentiation rules (2), (3), (7), and (5), we can conclude that polynomial functions are differentiable at every point z in the complex plane and rational functions are analytic throughout any domain D that contains no points at which the denominator is zero.

Theorem Polynomial and Rational Functions

(*i*) A polynomial function $p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$, where n is a nonnegative integer, is an entire function.

(*ii*) A rational function $f(z) = \dfrac{p(z)}{q(z)}$, where p and q are polynomial functions, is analytic in any domain D that contains no point z_0 for which $q(z_0) = 0$.

In general, a point z at which a complex function $w = f(z)$ fails to be analytic is called a **singular point** of f.

Theorem Differentiability Implies Continuity

If f is differentiable at a point z_0 in a domain D, then f is continuous at z_0.

Proof The limits $\lim_{z\to z_0} \dfrac{f(z)-f(z_0)}{z-z_0}$ and $\lim_{z\to z_0}(z-z_0)$ exist and equal $f'(z_0)$ and 0, respectively.

$$\begin{aligned}\lim_{z\to z_0}(f(z)-f(z_0)) &= \lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0}\cdot(z-z_0)\\ &= \lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0}\cdot \lim_{z\to z_0}(z-z_0) = f'(z_0)\cdot 0 = 0.\end{aligned}$$

From $\lim_{z\to z_0}(f(z)-f(z_0)) = 0$ we conclude that $\lim_{z\to z_0} f(z) = f(z_0)$.

Suppose f and g are functions that are analytic at a point z_0 and $f(z_0) = 0$, $g(z_0) = 0$, but $g'(z_0) \neq 0$. Then

$$\lim_{z \to z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}.$$

EXAMPLE 4 Using L'Hôpital's Rule

Compute $\displaystyle\lim_{z \to 2+i} \frac{z^2 - 4z + 5}{z^3 - z - 10i}$.

Solution If we identify $f(z) = z^2 - 4z + 5$ and $g(z) = z^3 - z - 10i$, you should verify that $f(2+i) = 0$ and $g(2+i) = 0$. The given limit has the indeterminate form $0/0$. Now since f and g are polynomial functions, both functions are necessarily analytic at $z_0 = 2 + i$. Using

$$f'(z) = 2z - 4, \quad g'(z) = 3z^2 - 1, \quad f'(2+i) = 2i, \quad g'(2+i) = 8 + 12i,$$

we see that

$$\lim_{z \to 2+i} \frac{z^2 - 4z + 5}{z^3 - z - 10i} = \frac{f'(2+i)}{g'(2+i)} = \frac{2i}{8 + 12i} = \frac{3}{26} + \frac{1}{13}i.$$

A Necessary Condition for Analyticity In the next theorem we see that if a function $f(z) = u(x,\ y) + iv(x,\ y)$ is differentiable at a point z, then the functions u and v must satisfy a pair of equations that relate their first-order partial derivatives.

Theorem Cauchy-Riemann Equations

Suppose $f(z) = u(x,\ y) + iv(x,\ y)$ is differentiable at a point $z = x + iy$. Then at z the first-order partial derivatives of u and v exist and satisfy the **Cauchy-Riemann equations**

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \qquad \text{and} \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \tag{1}$$

Proof The derivative of f at z is given by

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}. \tag{2}$$

By writing $f(z) = u(x,\ y) + iv(x,\ y)$ and $\Delta z = \Delta x + i\Delta y$, (2) becomes

$$f'(z) = \lim_{\Delta z \to 0} \frac{u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y) - u(x, y) - iv(x, y)}{\Delta x + i\Delta y}. \tag{3}$$

Since the limit (2) is assumed to exist, Δz can approach zero from any convenient direction. In particular, if we choose to let $\Delta z \to 0$ along a horizontal line, then $\Delta y = 0$ and $\Delta z = \Delta x$. We can then write (3) as

$$
\begin{aligned}
f'(z) &= \lim_{\Delta x \to 0} \frac{u(x+\Delta x, y) - u(x,y) + i\,[v(x+\Delta x, y) - v(x,y)]}{\Delta x} \\
&= \lim_{\Delta x \to 0} \frac{u(x+\Delta x, y) - u(x,y)}{\Delta x} + i \lim_{\Delta x \to 0} \frac{v(x+\Delta x, y) - v(x,y)}{\Delta x}.
\end{aligned} \tag{4}
$$

The existence of $f'(z)$ implies that each limit in (4) exists. These limits are the definitions of the first-order partial derivatives with respect to x of u and v, respectively. Hence, we have shown two things: both $\partial u/\partial x$ and $\partial v/\partial x$ exist at the point z, and that the derivative of f is

$$
f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}. \tag{5}
$$

We now let $\Delta z \to 0$ along a vertical line. With $\Delta x = 0$ and $\Delta z = i\Delta y$, (3) becomes

$$f'(z) = \lim_{\Delta y \to 0} \frac{u(x, y + \Delta y) - u(x, y)}{i\Delta y} + i \lim_{\Delta y \to 0} \frac{v(x, y + \Delta y) - v(x, y)}{i\Delta y}. \tag{6}$$

In this case (6) shows us that $\partial u/\partial y$ and $\partial v/\partial y$ exist at z and that

$$f'(z) = -i\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}. \tag{7}$$

By equating the real and imaginary parts of (5) and (7) we obtain the pair of equations in (1).

Because Theorem states that the Cauchy-Riemann equations (1) hold at z as a *necessary* consequence of f being differentiable at z, we cannot use the theorem to help us determine where f is differentiable. *But* it is important to realize that Theorem can tell us where a function f does not possess a derivative. If the equations in (1) are *not* satisfied at a point z, then f cannot be differentiable at z.

$f(z) = x + 4iy$ is not differentiable at any point z. If we identify $u = x$ and $v = 4y$, then $\partial u/\partial x = 1$, $\partial v/\partial y = 4$, $\partial u/\partial y = 0$, and $\partial v/\partial x = 0$. In view of

$$\frac{\partial u}{\partial x} = 1 \neq \frac{\partial v}{\partial y} = 4$$

the two equations in (1) cannot be simultaneously satisfied at any point z. In other words, f is nowhere differentiable.

if a complex function $f(z) = u(x, y)+iv(x, y)$ is analytic throughout a domain D, then the real functions u and v satisfy the Cauchy-Riemann equations (1) at every point in D.

EXAMPLE 1

The polynomial function $f(z) = z^2 + z$ is analytic for all z and can be written as $f(z) = x^2 - y^2 + x + i(2xy + y)$. Thus, $u(x, y) = x^2 - y^2 + x$ and $v(x, y) = 2xy + y$. For any point (x, y) in the complex plane we see that the Cauchy-Riemann equations are satisfied:

$$\frac{\partial u}{\partial x} = 2x + 1 = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -2y = -\frac{\partial v}{\partial x}.$$

Criterion for Non-analyticity

If the Cauchy-Riemann equations are not satisfied at every point z in a domain D, then the function $f(z) = u(x, y) + iv(x, y)$ cannot be analytic in D.

EXAMPLE 2 Using the Cauchy-Riemann Equations

Show that the complex function $f(z) = 2x^2 + y + i(y^2 - x)$ is not analytic at any point.

Solution We identify $u(x,\ y) = 2x^2 + y$ and $v(x,\ y) = y^2 - x$. From

$$\frac{\partial u}{\partial x} = 4x \quad \text{and} \quad \frac{\partial v}{\partial y} = 2y$$
$$\frac{\partial u}{\partial y} = 1 \quad \text{and} \quad \frac{\partial v}{\partial x} = -1 \tag{8}$$

we see that $\partial u/\partial y = -\partial v/\partial x$ but that the equality $\partial u/\partial x = \partial v/\partial y$ is satisfied only on the line $y = 2x$. However, for any point z on the line, there is no neighborhood or open disk about z in which f is differentiable at every point. We conclude that f is nowhere analytic.